

Def. Let K be a commutative ring with identity.

A map $D: M_{n \times n}(K) \rightarrow K$ is called n -linear if: for each $1 \leq i \leq n$,

$$D \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ c\alpha_i + \alpha'_i \\ \vdots \\ \alpha_n \end{pmatrix} = c \cdot D \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_i \\ \vdots \\ \alpha_n \end{pmatrix} + D \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha'_i \\ \vdots \\ \alpha_n \end{pmatrix}$$

when $c \in K$, $\alpha_1, \dots, \alpha_n, \alpha'_i$ are arbitrary row vectors.

Def. A n -linear map $D: M_{n \times n}(K) \rightarrow K$ is called Alternating if:

$D(A) = 0$ whenever two rows of A are equal.

Lemma. If n -linear map $D: M_{n \times n}(K) \rightarrow K$ is Alternating,

then for A' , obtained from A by interchanging two rows of A , $D(A') = -D(A)$.

Proof sketch. For odd n .

Def. Let K be a commutative ring with identity.

The unique map $D: M_n(K) \rightarrow K$ satisfying

$$n\text{-linear, alternating, } D(I) = 1$$

is called the determinant function.

Example. The determinant of 2 by 2 matrix is determined by: $\det A = A_{11}A_{22} - A_{12}A_{21}$.

Given $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = (A_{11}e_1 + A_{12}e_2, A_{21}e_1 + A_{22}e_2)$

Since D is n -linear,

$$\begin{aligned} D(A) &= D(A_{11}e_1 + A_{12}e_2, A_{21}e_1 + A_{22}e_2) \\ &= A_{11}D(e_1, A_{21}e_1 + A_{22}e_2) + A_{12}D(e_2, A_{21}e_1 + A_{22}e_2) \\ &= A_{11}A_{21}D(e_1, e_1) + A_{11}A_{22}D(e_1, e_2) + A_{12}A_{21}D(e_2, e_1) + A_{12}A_{22}D(e_2, e_2) \end{aligned}$$

Since D is alternating, $D(e_1, e_1) = D(e_2, e_2) = 0$ and $D(e_2, e_1) = -D(e_1, e_2) = -D(I) = -1$.

So,

$$= A_{11}A_{22} - A_{12}A_{21} \quad \square$$

A matrix $\text{adj } A = \begin{bmatrix} A_{22} & -A_{12} \\ -A_{21} & A_{11} \end{bmatrix}$ is called the classical adjoint of A .

Using the properties of $\det$, n -linear, alternating, $\det I = 1$, deduce the following:

a). $(\text{adj } A) \cdot A = A \cdot (\text{adj } A) = (\det A) \cdot I$.

b). $\det(\text{adj } A) = \det A$

c). $\text{adj}(A^t) = (\text{adj } A)^t$.

Proof. The first claim can be shown directly: $(\text{adj } A) \cdot A = \begin{bmatrix} A_{22}A_{11} - A_{12}A_{21} & 0 \\ 0 & -A_{21}A_{12} + A_{11}A_{22} \end{bmatrix}$

Similarly b) and c).

Given a $n \times n$ matrix A over a commutative ring with identity K , the i 'th row d_i is $\sum_{j=1}^n A_{ij} e_j$ for $1 \leq i \leq n$.

$$d_i = \sum_{j=1}^n A_{ij} \cdot e_j.$$

Therefore, for n -linear, alternating function D ,

$$\begin{aligned} D(A) &= D(d_1, d_2, \dots, d_n) = \sum_{j_1=1}^n A_{1,j_1} \cdot D(e_{j_1}, d_2, \dots, d_n) \\ &= \sum_{j_1=1}^n A_{1,j_1} \cdot \sum_{j_2=1}^n A_{2,j_2} \cdot D(e_{j_1}, e_{j_2}, d_3, \dots, d_n) \\ &\quad \vdots \\ &= \sum_{1 \leq j_1, \dots, j_n \leq n} A_{1,j_1} \cdot A_{2,j_2} \cdot \dots \cdot A_{n,j_n} \cdot D(e_{j_1}, \dots, e_{j_n}) \end{aligned}$$

Since D is alternating, if there are two of same indices, then $D(e_{j_1}, \dots, e_{j_n}) = 0$.

So, we can consider $(e_{j_1}, \dots, e_{j_n})$ as a permutation on n letters, moreover, if $D(I) = 1$, then $D(e_{j_1}, \dots, e_{j_n}) = \text{sgn}(j_1, \dots, j_n)$. In sum,

$$\det A = \sum_{\sigma \in S_n} \text{sgn } \sigma \cdot A_{1,\sigma(1)} \cdot \dots \cdot A_{n,\sigma(n)}.$$

✓ Invertibility of Matrix.

Lemma. Let K be a commutative Ring with identity, and let $A, B \in M_{n \times n}(K)$.

Then $\det AB = \det A \cdot \det B$.

Proof. Let B be a fixed $n \times n$ matrix over K . Define $D(A) = \det AB$.

Denote $\alpha_1, \dots, \alpha_n$ as i 'th rows of A . Then $D(\alpha_1, \dots, \alpha_n) = \det(\alpha_1 B, \dots, \alpha_n B)$.

Moreover, D is n -linear and alternating, since $\det$ is. So, $D(A) = \det A \cdot D(I) = \det A \cdot \det B$.

Def. Given $A \in M_{n \times n}(K)$, define the Classical adjoint of A as a matrix

$$(\text{adj } A)_{ji} \stackrel{\text{def}}{=} (-1)^{i+j} \det A(j|i).$$

where $A(j|i)$ is a $(n-1)$ by $(n-1)$ Matrix obtained by remove j 'th row and i 'th column.

Thm. Let K be a commutative Ring with identity.

Given $A \in M_{n \times n}(K)$,

$$(\text{adj } A) \cdot A = (\det A) \cdot I.$$

Proof.

